WORKED SOLUTION

STEP III 2023, Question 3

Pure •

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Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

Measured from a, the three points are $sb\mathrm{i}$, $-sb\mathrm{i}$ and b. Multiplying by i turns a vector through a right angle, so a is the midpoint of the base and b points along the axis of symmetry. With that picture, part (ii) is a matter of Vieta's formulas: the roots sum to zero, which pins b down in terms of a, and then a cancels from p^3/q^2. Parts (iii) and (iv) turn the result into a one-variable function of $x = s^2 > 0$, whose graph shows how small it can be.

Solution

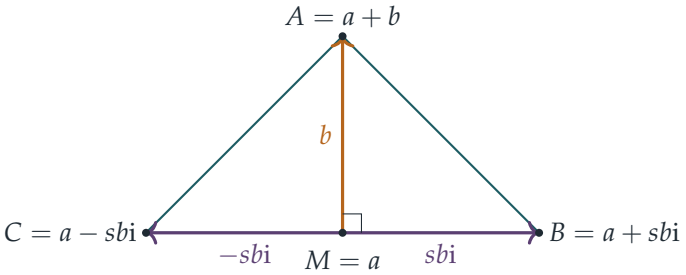
Part (i)

Distances from $a + b$ to the other two points are

$$|(a + sb\mathrm{i}) - (a + b)| = |b|\,|s\mathrm{i} - 1| = |b|\sqrt{s^2 + 1},$$
$$|(a - sb\mathrm{i}) - (a + b)| = |b|\,|-s\mathrm{i} - 1| = |b|\sqrt{s^2 + 1}.$$

These are equal. We still need the three points to form a genuine triangle. Relative to a they are $sb\mathrm{i}$, $-sb\mathrm{i}$ and b. The first two are distinct (as $s > 0$ and $b \neq 0$) and lie on the line through a in the direction of $b\mathrm{i}$. Since $b\mathrm{i}$ is b rotated through a right angle, the third point $a + b$ lies off that line, because $b \neq 0$. So the points form an isosceles triangle with apex $a + b$, base from $a - sb\mathrm{i}$ to $a + sb\mathrm{i}$, and base midpoint a.

Recovering a, b, s. Let the triangle have apex A (the vertex where the two equal sides meet; in an equilateral triangle any vertex will do) and other vertices B, C. Let M be the midpoint of BC.



- Take $a = M = \frac{1}{2}(B + C)$.
- Take $b = A - M$. This is non-zero, as A is not on BC.
- Since $AB = AC$, A lies on the perpendicular bisector of BC, so $MB \perp MA$. Hence $B - M$ is b turned through a right angle and rescaled: $B - M = \lambda b\mathrm{i}$ for some real $\lambda \neq 0$. If $\lambda < 0$, swap the labels B and C, which changes the sign of λ. Then $s = \lambda > 0$ is
$$s = \frac{MB}{MA} = \frac{\frac{1}{2}BC}{\text{height of the triangle}}.$$

Then $a + b = A$, $a + sb\mathrm{i} = B$ and $a - sb\mathrm{i} = M - (B - M) = C$, as required.

Part (ii)

By part (i), we can label the roots $a + sb\mathrm{i}$, $a - sb\mathrm{i}$, $a + b$ with $b \neq 0$, $s > 0$. The cubic has no z^2 term, so the roots sum to zero:

$$3a + b = 0, \qquad \text{so } b = -3a,$$

and $a \neq 0$ because $b \neq 0$. The roots are therefore

$$a(1 - 3s\mathrm{i}), \qquad a(1 + 3s\mathrm{i}), \qquad -2a.$$

The first two have product $a^2(1 + 9s^2)$ and sum $2a$. So, by Vieta's formulas,

$$p = a^2(1 + 9s^2) + (-2a)(2a) = a^2(9s^2 - 3) = 3a^2(3s^2 - 1),$$
$$q = -(\text{product of roots}) = -(-2a)\,a^2(1 + 9s^2) = 2a^3(9s^2 + 1).$$

Here $q \neq 0$ since $a \neq 0$, so we may divide:

$$\frac{p^3}{q^2} = \frac{27a^6(3s^2 - 1)^3}{4a^6(9s^2 + 1)^2} = \frac{27(3s^2 - 1)^3}{4(9s^2 + 1)^2},$$

with s a non-zero (indeed positive) real number, as required.

Part (iii)

Let $y = \dfrac{(3x-1)^3}{(9x+1)^2}$, defined for $x \neq -\frac{1}{9}$. By the quotient rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{9(3x-1)^2(9x+1)^2 - (3x-1)^3 \cdot 18(9x+1)}{(9x+1)^4}$$
$$= \frac{9(3x-1)^2\big[(9x+1) - 2(3x-1)\big]}{(9x+1)^3} = \frac{27(3x-1)^2(x+1)}{(9x+1)^3}.$$

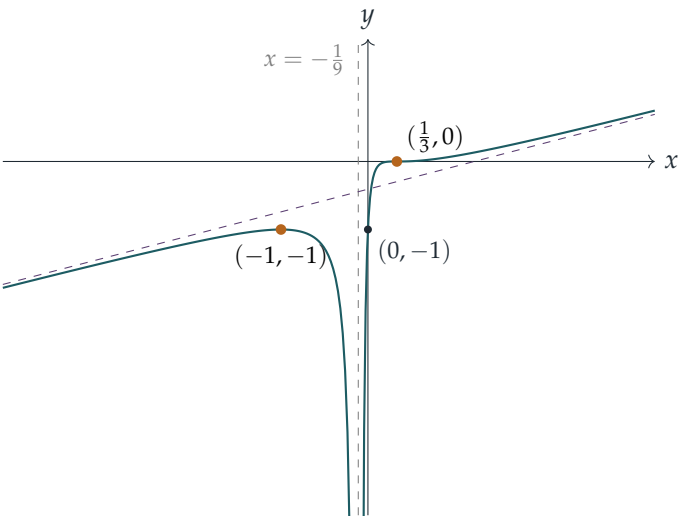
So the stationary points are at $x = -1$ and $x = \frac{1}{3}$.

- At $x = -1$: $y = (-4)^3/(-8)^2 = -1$. For $x < -1$, the factors $x+1$ and $(9x+1)^3$ are both negative, so $y' > 0$. For $-1 < x < -\frac{1}{9}$, only $(9x+1)^3$ is negative, so $y' < 0$. So $(-1,-1)$ is a **local maximum**.
- At $x = \frac{1}{3}$: $y = 0$. On both sides (for $x > -\frac{1}{9}$) every factor except $(3x-1)^2$ is positive, and $(3x-1)^2 > 0$ for $x \neq \frac{1}{3}$. So $y' > 0$ on either side, and $(\frac{1}{3}, 0)$ is a **stationary point of inflection**.

In particular, y is strictly increasing on $x > -\frac{1}{9}$, since $y' > 0$ there except at the single point $x = \frac{1}{3}$.

Other features:

- Vertical asymptote $x = -\frac{1}{9}$. There the numerator tends to $(-\frac{4}{3})^3 < 0$ and the denominator to 0^+, so $y \to -\infty$ from both sides.
- Dividing $27x^3 - 27x^2 + 9x - 1$ by $81x^2 + 18x + 1$ gives quotient $\frac{1}{3}x - \frac{11}{27}$, so $y = \frac{1}{3}x - \frac{11}{27}$ is an oblique asymptote as $x \to \pm\infty$.
- The curve meets the axes at $(0,-1)$ and $(\frac{1}{3}, 0)$.



The dashed line is the oblique asymptote $y = \frac{1}{3}x - \frac{11}{27}$. The left branch never rises above -1. The right branch increases from $-\infty$ through every real value.

Part (iv)

By part (ii), with the real number $s \neq 0$ found there and $x = s^2$,

$$\frac{p^3}{q^2} = \frac{27}{4} \cdot \frac{(3x-1)^3}{(9x+1)^2}.$$

Since s is real, the right-hand side is real, and therefore so is p^3/q^2.

Ask yourself The graph goes down to $-\infty$, so why can't p^3/q^2 be very negative?

Because $x = s^2$ is not just any real number: s is real and non-zero, so $x > 0$. The values of x we care about therefore lie on the right branch, where (from part (iii)) y is strictly increasing. Hence, for $x > 0$,

$$\frac{(3x-1)^3}{(9x+1)^2} > \frac{(0-1)^3}{(0+1)^2} = -1,$$

and so

$$\frac{p^3}{q^2} > -\frac{27}{4}.$$

The inequality is strict because $x = 0$ (that is, $s = 0$) is excluded. It would make the triangle collapse to a line segment.

Checks

- Equilateral triangle: the roots of $z^3 - 1 = 0$ form an equilateral triangle. Here $p = 0$, and the formula gives $p = 0$ exactly when $s^2 = \frac{1}{3}$, i.e. when the half-base is $\frac{1}{\sqrt{3}}$ times the height, which is the equilateral shape.
- A concrete case: $a = 1$, $s = 1$ gives roots $1 \pm 3\mathrm{i}$ and -2, so $z^3 + 6z + 20 = 0$. Then $p^3/q^2 = 216/400 = \frac{27}{50}$, and the formula gives $\frac{27 \cdot 8}{4 \cdot 100} = \frac{27}{50}$.
- As $s \to 0$ the triangle flattens towards a double root at a, and $p^3/q^2 \to -\frac{27}{4}$: $4p^3 + 27q^2 = 0$ is exactly the condition for a repeated root. For $s = 10^{-4}$ the value is about -6.749998.
- Random isosceles triangles with centroid at the origin were checked by computer: p^3/q^2 was always real and greater than -6.75.

Common slips

- Treating a and b as real numbers in part (i). They are complex, so the triangle can sit at any angle.

- Getting the sign of q wrong: for a monic cubic, q is *minus* the product of the roots.
- In part (iv), using the whole graph and forgetting that $x = s^2 > 0$, which is what keeps us on the right branch.

Beyond STEP

The quantity $-(4p^3 + 27q^2)$ is the discriminant of the cubic, and p^3/q^2 is a shape invariant: it is unchanged when the roots are rotated or scaled about their centroid (replacing z by kz multiplies p by k^2 and q by k^3), so it records only the *shape* of the triangle of roots. Invariants like this are the starting point of classifying polynomials up to change of variable. Needham's *Visual Complex Analysis* is a beautiful guide to thinking geometrically with complex numbers in the way this question asks.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.